

SIMATS ENGINEERING
ITA1408-ETHICAL HACKING

CO2-ASSESSMENT TOOL-O2

Social Engineering Simulation

Name:A.Sai Joshitha[192424220]

Question 1

As a Security Analyst, perform an OSINT-based email enumeration assessment on the target domain tesla.com using theHarvester in Kali Linux. Collect publicly available email addresses, subdomains, hosts, and IP addresses associated with the organization. Categorize the discovered email addresses into likely roles such as Support, Sales, Human Resources, Administration, and Technical Teams. Analyze how the gathered information could assist an attacker during the reconnaissance phase and explain how it could be used for spear-phishing, credential harvesting, and impersonation attacks. Finally, recommend at least five countermeasures to reduce information leakage and social engineering risks, and submit a report with screenshots, commands executed, findings, risk analysis, and recommendations. (20 Marks)

Aim

To perform an OSINT-based email enumeration assessment on the target domain tesla.com using theHarvester in Kali Linux.

Objective

- Collect publicly available email addresses.
- Identify subdomains and hosts.
- Discover IP addresses.
- Categorize email addresses based on departments.
- Analyze social engineering risks.
- Recommend security countermeasures.

Tools Required

- Kali Linux
- theHarvester
- Terminal
- Internet connection

Command Used

Command 1 : theHarvester -d tesla.com -b rapiddns

Procedure

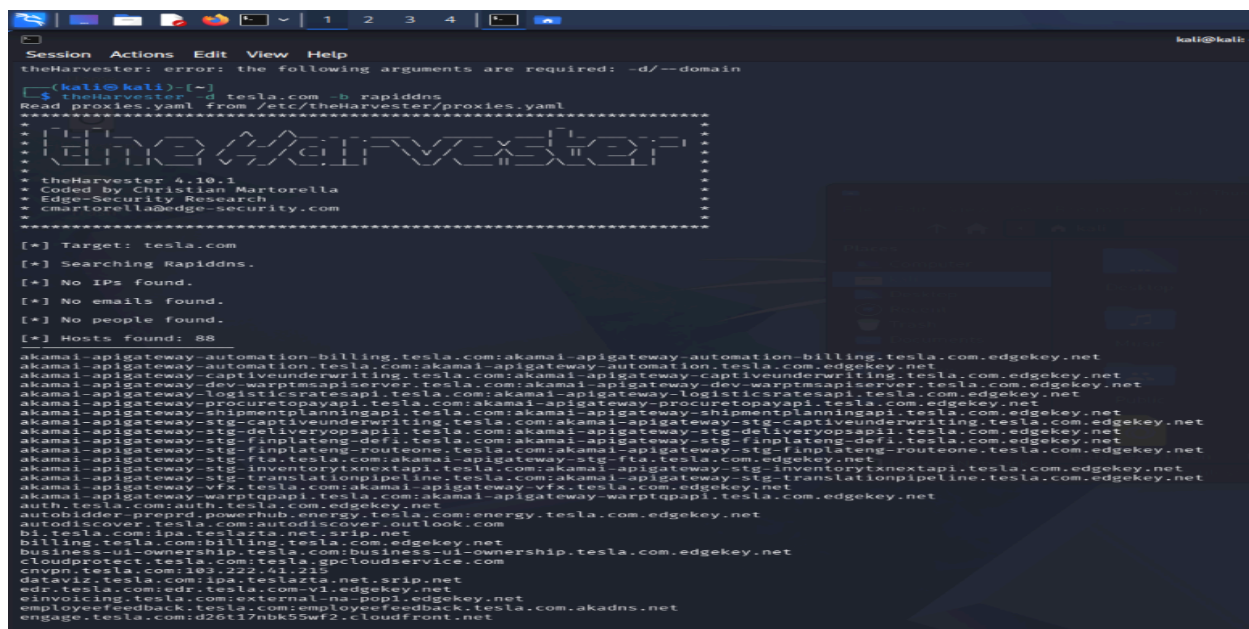
Step 1

Open Kali Linux Terminal.

Step 2

Run the command:

theHarvester -d tesla.com -b rapiddns



```
Session Actions Edit View Help
theHarvester: error: the following arguments are required: -d/--domain

(kali@kali)~$ theHarvester -d tesla.com -b rapiddns
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
[*] Searching RapidDns.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 88
akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-dev-warptmsapiserver.tesla.com:akamai-apigateway-dev-warptmsapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticsratesapi.tesla.com:akamai-apigateway-logisticsratesapi.tesla.com.edgekey.net
akamai-apigateway-procuretopayapi.tesla.com:akamai-apigateway-procuretopayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi.tesla.com:akamai-apigateway-stg-deliveryopsapi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-routeone.tesla.com:akamai-apigateway-stg-finplateng-routeone.tesla.com.edgekey.net
akamai-apigateway-stg-fts.tesla.com:akamai-apigateway-stg-fts.tesla.com.edgekey.net
akamai-apigateway-stg-inventorytnextapi.tesla.com:akamai-apigateway-stg-inventorytnextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warptqapi.tesla.com:akamai-apigateway-warptqapi.tesla.com.edgekey.net
auth.tesla.com:auth.tesla.com.edgekey.net
autobidder-preprod.powerhub.energy.tesla.com:energy.tesla.com.edgekey.net
autodiscover.tesla.com:autodiscover.outlook.com
bi.tesla.com:ipa.tesla.net.srip.net
billing.tesla.com:billing.tesla.com.edgekey.net
business-ai-ownership.tesla.com:business-ai-ownership.tesla.com.edgekey.net
cloudprotect.tesla.com:tesla.gpcloudservice.com
cnvnp.tesla.com:101.222.41.215
dataviz.tesla.com:ipa.tesla.net.srip.net
edr.tesla.com:edr.tesla.com.v1.edgekey.net
einvoiceing.tesla.com:external-na-pop1.edgekey.net
employeefeedback.tesla.com:employeefeedback.tesla.com.akadns.net
engage.tesla.com:d26t17nbk55wf2.cloudfront.net
```

```
Session Actions Edit View Help
errlog.tesla.com:errlog.tesla.com.edgekey.net
forums.tesla.com:site-6030136.onvanilla.net
grid.tesla.com:grid.tesla.com.edgekey.net
hermes-ap.eng.vn.cloud.tesla.com:usw-eng5x-hermes-npuaph-prd.teslafleet.akadns.net
image-emails.tesla.com:image-emails.tesla.com.edgekey.net
image.email.tesla.com:image.s7.exacttarget.com.edgesuite.net
links.tesla.com:links.tesla.com.edgekey.net
livestreamapi.tesla.com:livestreamapi.tesla.com.edgekey.net
mfs-supplier.tesla.com:mfs-supplier.tesla.com.edgekey.net
mobile.tesla.com:mobile.tesla.com.edgekey.net
mta.email.tesla.com:13.111.14.190
npuhe-eng.euw1.vn.cloud.tesla.com:npuhe-eng-c-286de95ea00c5e2a.elb.eu-west-1.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-a-c517f248c251dc44.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-b-979133c19331427b.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-c-ef07d5b3ed829c9b.elb.us-west-2.amazonaws.com
npute-eng.usw.vn.cloud.tesla.com:usw-eng5a-fleet-npute-prd.teslafleet.akadns.net
npute-eng.usw.vn.cloud.tesla.com:usw-eng5c-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw2.vn.cloud.tesla.com:npute-prd-a-ea25687038eacb2e.elb.us-west-2.amazonaws.com
npuv-ap.prd.euw1.vn.cloud.tesla.com:npuv-prd.euw1.vn.cloud.tesla.com
npuv-eng.euw1.vn.cloud.tesla.com:npuv-eng-a-38adfa93f5795d7c.elb.eu-west-1.amazonaws.com
npuv-eng.usw.vn.cloud.tesla.com:usw-eng5d-fleet-npuv-prd.teslafleet.akadns.net
npuv-eng.usw2.vn.cloud.tesla.com:npuv-eng-a-97483e916dd01bb4.elb.us-west-2.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-a-56bc286e6fb51aaf.elb.eu-west-1.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-c-04d07fce0e0dc20b.elb.eu-west-1.amazonaws.com
npuv-prd.usw.vn.cloud.tesla.com:usw-prd5c-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd.usw2.vn.cloud.tesla.com:npuv-prd-c-a2aa85266360f4cf.elb.us-west-2.amazonaws.com
origin-bolt.tesla.com:clsmdcgyp.tesla.com.akadns.net
origin-courses.tesla.com:origin-courses.tesla.com.akadns.net
origin-edr.tesla.com:clsgenp.tesla.com.akadns.net
origin-einvoicing.tesla.com:warp-einvoicing.tesla.com.akadns.net
origin-finops.tesla.com:clsmdcfinp.tesla.com.akadns.net
origin-livestream.tesla.com:k8clsgen.tesla.com.akadns.net
origin-location-services-prd.tesla.com:clsmdcvehp.tesla.com.akadns.net
origin-static-assets-profile-settings.tesla.com:clsmdcgenp.tesla.com.akadns.net
origin-tcc-gw-stg.tesla.com:edgeservices.tesla.com.akadns.net
origin-warp-akamai.tesla.com:199.66.9.77
partners.tesla.com:partners.tesla.com.edgekey.net
sales-prd.tesla.com:external-na-pop1.edgekey.net
shop.tesla.com:shop.tesla.com.edgekey.net
static-assets-pay.tesla.com:static-assets-pay.tesla.com-v1.edgekey.net
static-assets.tesla.com:static-assets.tesla.com.edgekey.net
stg-uc-twilio-bpm-webhook.tesla.com:external-na-pop1.edgekey.net
suppliers.tesla.com:suppliers.tesla.com.edgekey.net
toolbox.tesla.com:toolbox.tesla.com.edgekey.net
trt.tesla.com:external-na-pop1.edgekey.net
ul.engage.tesla.com:d1e60lajz6qsud.cloudfront.net
url2547.tesla.com:sendgrid.net
url5196.tesla.com:sendgrid.net
url7051.tesla.com:url7051.tesla.com.edgekey.net
```

Command 2 : theHarvester -d tesla.com -b rapiddns -n

```
Session Actions Edit View Help
links.tesla.com:links.tesla.com.edgekey.net
livestreamapi.tesla.com:livestreamapi.tesla.com.edgekey.net
mfs-supplier.tesla.com:mfs-supplier.tesla.com.edgekey.net
mobile.tesla.com:mobile.tesla.com.edgekey.net
mta.email.tesla.com:13.111.14.190
npuhe-eng.euw1.vn.cloud.tesla.com:npuhe-eng-c-286de95ea00c5e2a.elb.eu-west-1.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-a-c517f248c251dc44.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-b-979133c19331427b.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-c-ef07d5b3ed829c9b.elb.us-west-2.amazonaws.com
npute-eng.usw.vn.cloud.tesla.com:usw-eng5a-fleet-npute-prd.teslafleet.akadns.net
npute-eng.usw.vn.cloud.tesla.com:usw-eng5c-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw2.vn.cloud.tesla.com:npute-prd-a-ea25687038eacb2e.elb.us-west-2.amazonaws.com
npuv-ap.prd.euw1.vn.cloud.tesla.com:npuv-prd.euw1.vn.cloud.tesla.com
npuv-eng.euw1.vn.cloud.tesla.com:npuv-eng-a-38adfa93f5795d7c.elb.eu-west-1.amazonaws.com
npuv-eng.usw2.vn.cloud.tesla.com:npuv-eng-a-97483e916dd01bb4.elb.us-west-2.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-a-56bc286e6fb51aaf.elb.eu-west-1.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-c-04d07fce0e0dc20b.elb.eu-west-1.amazonaws.com
npuv-prd.usw.vn.cloud.tesla.com:usw-prd5c-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd.usw2.vn.cloud.tesla.com:npuv-prd-c-a2aa85266360f4cf.elb.us-west-2.amazonaws.com
origin-bolt.tesla.com:clsmdcgyp.tesla.com.akadns.net
origin-courses.tesla.com:origin-courses.tesla.com.akadns.net
origin-edr.tesla.com:clsgenp.tesla.com.akadns.net
origin-einvoicing.tesla.com:warp-einvoicing.tesla.com.akadns.net
origin-finops.tesla.com:clsmdcfinp.tesla.com.akadns.net
origin-livestream.tesla.com:k8clsgen.tesla.com.akadns.net
origin-location-services-prd.tesla.com:clsmdcvehp.tesla.com.akadns.net
origin-static-assets-profile-settings.tesla.com:clsmdcgenp.tesla.com.akadns.net
origin-tcc-gw-stg.tesla.com:edgeservices.tesla.com.akadns.net
origin-warp-akamai.tesla.com:199.66.9.77
partners.tesla.com:partners.tesla.com.edgekey.net
sales-prd.tesla.com:external-na-pop1.edgekey.net
shop.tesla.com:shop.tesla.com.edgekey.net
static-assets-pay.tesla.com:static-assets-pay.tesla.com-v1.edgekey.net
static-assets.tesla.com:static-assets.tesla.com.edgekey.net
stg-uc-twilio-bpm-webhook.tesla.com:external-na-pop1.edgekey.net
suppliers.tesla.com:suppliers.tesla.com.edgekey.net
toolbox.tesla.com:toolbox.tesla.com.edgekey.net
trt.tesla.com:external-na-pop1.edgekey.net
ul.engage.tesla.com:d1e60lajz6qsud.cloudfront.net
url2547.tesla.com:sendgrid.net
url5196.tesla.com:sendgrid.net
url7051.tesla.com:url7051.tesla.com.edgekey.net

[*] Starting active queries for DNSLookup.
[*] Hosts found after reverse lookup (in target domain):
```

```
kali@kali: ~
Session Actions Edit View Help
url7051.tesla.com:url7051.tesla.com.edgekey.net

(kali@kali)~$ theHarvester -d tesla.com -b rapiddns -n
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
[*] Searching RapidDns.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 88

akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-dev-warptmsapiserver.tesla.com:akamai-apigateway-dev-warptmsapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticsratesapi.tesla.com:akamai-apigateway-logisticsratesapi.tesla.com.edgekey.net
akamai-apigateway-procuretopayapi.tesla.com:akamai-apigateway-procuretopayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi1.tesla.com:akamai-apigateway-stg-deliveryopsapi1.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-defi.tesla.com:akamai-apigateway-stg-finplateng-defi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-routeone.tesla.com:akamai-apigateway-stg-finplateng-routeone.tesla.com.edgekey.net
akamai-apigateway-stg-fts.tesla.com:akamai-apigateway-stg-fts.tesla.com.edgekey.net
akamai-apigateway-stg-inventorytxnextapi.tesla.com:akamai-apigateway-stg-inventorytxnextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warptqpapi.tesla.com:akamai-apigateway-warptqpapi.tesla.com.edgekey.net
auth.tesla.com:auth.tesla.com.edgekey.net
autobidder-preprd.powerhub.energy.tesla.com:energy.tesla.com.edgekey.net
autodiscover.tesla.com:autodiscover.outlook.com
```

Step 3

Collect the following information:

- Email addresses
- Subdomains
- Hosts
- IP addresses

Step 4

Categorize the email addresses into:

- Support Team
- Sales Team
- Human Resources
- Administration Team
- Technical Team

Risk Analysis

The collected information can be used for:

- Phishing attacks
- Credential theft
- Employee impersonation
- Password attacks
- Social engineering attacks

Countermeasures

1. Enable multi-factor authentication.
2. Hide sensitive employee emails.
3. Conduct security awareness training.
4. Use email filtering.
5. Monitor suspicious activities.

Conclusion

OSINT tools can expose sensitive information that attackers may exploit. Organizations must protect their public data and train employees against social engineering attacks.

Question 2

As a cybersecurity awareness consultant, conduct a phishing awareness simulation using GoPhish and dummy employee accounts provided in the lab. Create an email template, launch a campaign, monitor user interactions, and generate a report showing delivery rate, email open rate, and click rate. Based on the results, provide recommendations to improve user awareness against phishing attacks. (20 Marks)

Aim

To conduct a phishing awareness simulation using GoPhish and analyze user interactions.

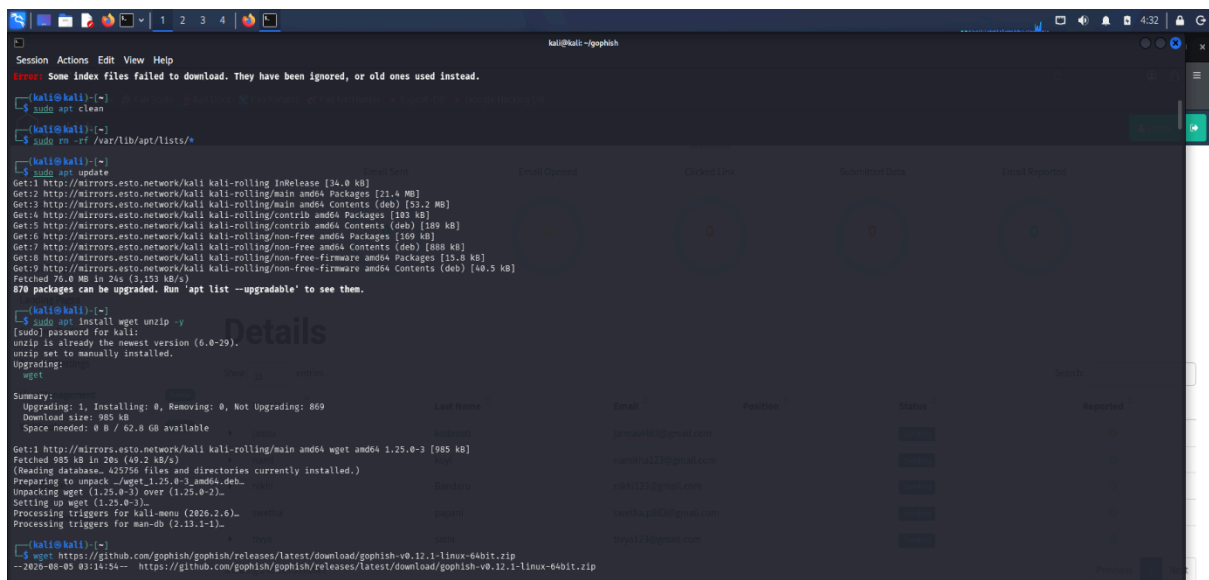
Objective

- Create a phishing email template.
- Launch a phishing campaign.
- Monitor employee responses.
- Measure delivery, open, and click rates.
- Improve employee awareness.

Tools Required

- Kali Linux
- GoPhish
- Web browser

Command 1 : sudo apt update



```
kali@kali: ~ - ssh
Session Actions Edit View Help
Error: Some index files failed to download. They have been ignored, or old ones used instead.

(kali@kali)~$ sudo apt clean
(kali@kali)~$ sudo rm -rf /var/lib/apt/lists/*
(kali@kali)~$ sudo apt update
Get:1 http://mirrors.esto.network/kali kali-rolling/InRelease [34.0 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get:4 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [102 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [189 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [169 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [888 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [15.0 kB]
Get:9 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.5 kB]
Fetched 76.0 MB in 24s (3.152 kB/s)
870 packages can be upgraded. Run 'apt list --upgradable' to see them.
(kali@kali)~$ sudo apt install wget unzip -y
[sudo] password for kali:
wget is already the newest version (6.0-29).
unzip is already the newest version (6.0-29).
unzip set to manually installed.
Upgrading:
  wget
Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 869
  Download size: 985 kB
  Space needed: 0 B / 62.8 GB available
Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 wget amd64 1.25.0-3 [985 kB]
Fetched 985 kB in 20s (49.2 kB/s)
(Reading database... 422756 files and directories currently installed.)
Preparing to unpack ./wget_1.25.0-3_amd64.deb...
Unpacking wget (1.25.0-3) over (1.25.0-2)...
Setting up wget (1.25.0-3)...
Processing triggers for kali-menu (2020.2.6)...
Processing triggers for man-db (2.13.1-1)...
(kali@kali)~$ wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
--2020-08-05 03:14:54-- https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
```

<https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip>

Command 4 : cd gophish

Command 5 : `sudo ./gophish`



Open the GoPhish dashboard.

Step 2

Create employee groups.

Step 3

Create an email template.

Step 4

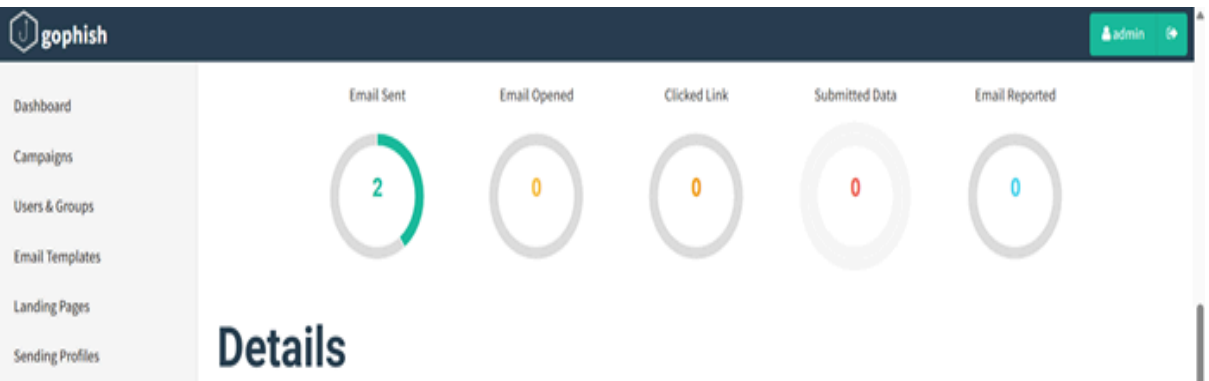
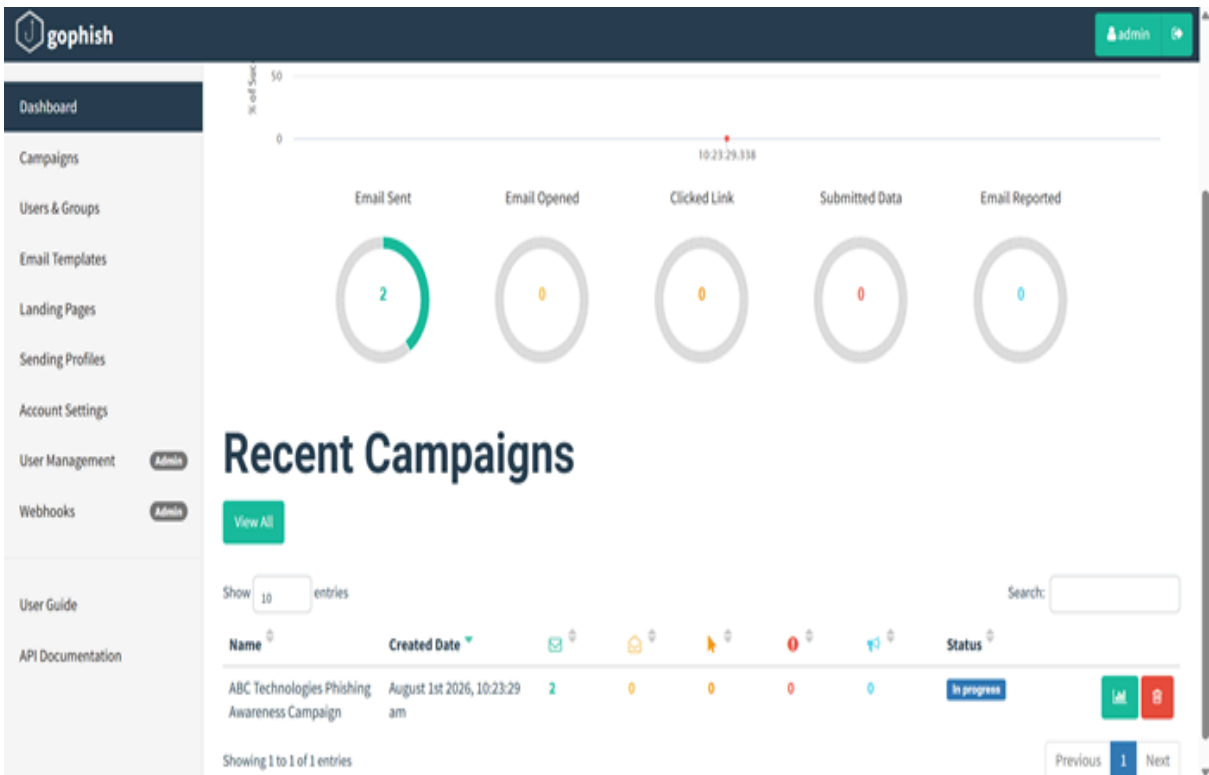
Create a landing page.

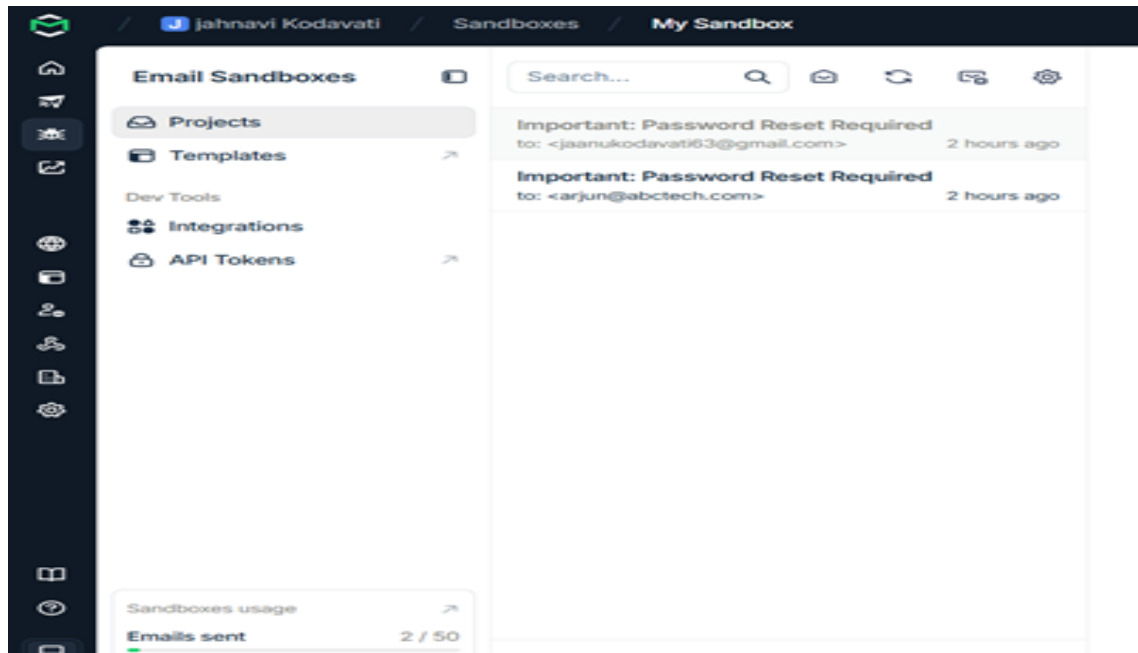
Step 5

Launch the campaign.

Step 6

Monitor the results.





Recommendations

- Verify sender emails.
- Avoid unknown links.
- Use strong passwords.
- Enable multi-factor authentication.
- Report suspicious emails.

Conclusion

Phishing simulations help organizations improve employee awareness and reduce cyber risks.

Question 3

Using Nmap Aggressive Scan (-A) on scanme.nmap.org, gather detailed information about the target, including operating system details, open ports, and service versions. Assume you are an attacker and describe how the collected information could be leveraged to conduct a social engineering attack against the IT support team. Prepare a report explaining the attack methodology and defensive recommendations. (10 Marks)

Aim

To perform an aggressive scan on the target scanme.nmap.org using Nmap.

Objective

- Identify open ports.
- Detect services.
- Find operating system details.
- Analyze security risks.
- Recommend defensive measures.

Tool Required

- Kali Linux
- Nmap

Command Used

Command 1 : nmap -A scanme.nmap.org
--

Procedure

Step 1

Open Kali Linux.

Step 2

Run the command:

nmap -A scanme.nmap.org

```
kali@kali:~$ nmap -sC -sV 10.10.10.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 06:09 -0400
Nmap scan report for scanner.nmap.org (45.33.32.156)
Host is up (0.0015s latency).
Other addresses for scanner.nmap.org (not scanned): 2000:3C01::F83C:93FF:F018:D02F
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
135/tcp   closed rdp
137/tcp   closed domain
138/tcp   open  http      Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed mitl-dev
143/tcp   closed imap
144/tcp   closed https
193/tcp   closed imap
2309/tcp  closed ms-mdt-server
2800/tcp  closed pfs
2801/tcp  open  sip
2802/tcp  open  sip-tls
2803/tcp  closed http-proxy
2804/tcp  closed blackice-icecap
2805/tcp  closed unknown
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (80%), Slirp (80%), QEMU (80%)
OS CPE: cpe:/o:oracle-virtualbox:qemu/dmcc/dmcc:slirp/qemu:qemu
Aggressive OS guesses: AT&T BCM210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (80%), QEMU user mode network gateway (80%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 2.76 ms scanner.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 123.74 seconds
kali@kali:~$
```

Step 3

Collect information such as:

- Open ports
- Service versions
- Operating system details

Social Engineering Scenario

An attacker can:

- Impersonate the IT team.
- Send fake emails.
- Ask employees for credentials.
- Exploit exposed services.

Defensive Recommendations

1. Close unnecessary ports.
2. Use firewalls.
3. Update software regularly.
4. Train employees.
5. Use strong passwords.

Conclusion

Nmap helps security professionals identify vulnerabilities and improve the security of computer networks.